

Math 170E
Week 5

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1 Continuous Random Variables & Poisson Distribution

Example 1. On average, how many rolls we need to throw a fair dice to get all 6 outcomes?

- (A)6 (B)36 (C)12 (D)14.7 (E) $\frac{144}{7}$

$$\frac{6}{1} + \frac{6}{2} + \frac{6}{3} + \frac{6}{4} + \frac{6}{5} + \frac{6}{6} = 14.7$$

Example 2. (PSI 2.7.5) Flaws in a certain type of drapery material appear on the average of one in 150 square feet. If we assume a Poisson distribution, approximate the probability of at most one flaw appearing in 225 square feet.

- (A)0.06 (B)0.12 (C)0.33 (D)0.56 (E)0.67

$$X = \text{Poisson}\left(\frac{225}{150}\right) = \text{Poisson}(1.5)$$
$$P(X \leq 1) = P(X=0) + P(X=1) = \lambda^0 \frac{e^{-\lambda}}{0!} + \lambda^1 \frac{e^{-\lambda}}{1!} = \frac{5}{2} e^{-\frac{3}{2}} \approx 0.66$$

$\lambda = \frac{3}{2}$

Example 3. (PSI 3.1.15) The life X (in years) of a voltage regulator of a car has the pdf

$$f(x) = \frac{3x^2}{7^3} e^{-(x/7)^3}$$

for $0 < x < \infty$.

1. What is the probability that this regulator will last at least 7 years?

- (A) e^{-1} (B) e^{-7} (C) $e^{-1/7}$ (D) $\frac{1}{7}$ (E) $\frac{316}{343}$

2. Given that it has lasted at least 7 years, what is the conditional probability that it will last at least another 3.5 years?

- (A) e^{-1} (B) $e^{-19/8}$ (C) $e^{-27/8}$ (D) $\frac{7}{8}$ (E) $\frac{27}{343}$

$$X \sim \text{Geo}\left(\frac{0.01}{p}\right) \quad E[X] = \frac{1}{p} = 100, \quad \text{Var}[X] = \frac{1-p}{p^2} = 9900$$

Example 4. (Old MT problem) Let us consider an unfair coin which tosses head and tail with probability 0.01 and 0.99, respectively.

1. Toss this coin independently, and let X be the first time that head appears. Compute the mean and variance of X .
2. Toss this coin for 500 times independently. Use a Poisson limit theorem to compute (approximately) the probability that exactly 5 many head appears.

$$Y = \# \text{ of heads appeared in } 500$$

$$Y \sim \text{Pois}\left(500 \cdot \frac{1}{100}\right) = \text{Pois}(5) \text{ approximately}$$

$$P(Y=5) = \frac{5^5 e^{-5}}{5!}$$

Example 5. (Old Quiz Problem) Let X be a continuous random variable with uniform distribution $U([3, 4])$.

1. Compute $\mu = EX$

(A)3 (B)3.5 (C)4 (D)7 (E)49

2. Compute $M(t)$

(A) t^7 (B) $e^{7t/2}$ (C) $\frac{e^{7t/2}}{12}$ (D) $\frac{e^{4t} - e^{3t}}{12}$ (E) $\frac{e^{4t} - e^{3t}}{t}$

3. Compute $\sigma^2 = \text{Var}(X)$

(A)1 (B)12 (C)7 (D) $\frac{1}{7}$ (E) $\frac{1}{12}$

Example 6. (Old MT problem) Suppose that a coin is not fair so that the probability of obtaining a head is $p \in (0, 1)$.

1. On average, how many flips are needed to obtain a head? $\frac{1}{p}$
2. Find the probability the first time obtaining a head is an even number. Your final answer must not be an infinite series.

$$X \sim \text{Geo}(p)$$

$$P(X = \text{even}) = \sum_{k=0}^{\infty} P(X = 2k) = \underbrace{(1-p)p}_{a} + \underbrace{(1-p)^3 p}_{ar} + (1-p)^5 p + \dots$$

$r = (1-p)^2$

$$= \frac{a}{1-r} = \frac{P(\text{H})}{1-(1-p)^2} = \frac{1-p}{2-p}$$